

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. HONORS DEGREE EXAMINATIONS - NOVEMBER 2025

Deep Learning [ECE_5101H]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Find the intercept and slope of the following using linear regression model using matrix form (4)

A)

X	Y
1	5
2	7
3	9
4	11
5	13
6	15

You have given the following inverse($X^T X$) =

$$I = \begin{bmatrix} 0.8667 & -0.200 \\ -0.200 & 0.0571 \end{bmatrix}$$

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) Distinguish between supervised and unsupervised learning, providing one application of each (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Sketch a decision surface for a two-dimensional linear classification problem and label the margin. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) With a diagram, explain the functioning of a multilayer perceptron (MLP). Provide an example of its application in real life. (4)

- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Define autoencoders. Describe how their loss functions guide the training process. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) What is hyperparameters in Deep neural network ? Discuss various hyperparameter tuning methods. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Describe residual networks (ResNets). How do skip connections help with vanishing gradient problems? Support your answer with equations or diagrams where possible.. (4)

- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Discuss early stopping as a regularization technique in deep learning. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- C) What is the importance of skip connections in deep neural networks?. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) List two use cases for LSTM networks outside text processing, explaining why LSTM is suitable. (4)
A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Write a basic numerical example to show how object detection is evaluated using bounding box predictions. (3)
Course Outcome: "CO4" / Cognitive Level: "Understand"
- C) With the help of equations, explain how a variational autoencoder learns to reconstruct input data (3)
Course Outcome: "CO4" / Cognitive Level: "Understand"
- 5) Suppose you have a set of data points; illustrate how decision surfaces change when you modify the decision boundary (4)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Justify with your answer that why autoencoder are more suitable than Principal component analysis in Dimensionality reduction. (3)
Course Outcome: "CO4" / Cognitive Level: "Analysis"
- C) Discuss the bias and Variance tradeoff. (3)
Course Outcome: "CO1" / Cognitive Level: "Analysis"

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Event Name	Subject
058-END SEMSTER-ODD-2025 ECE_5101H:Deep Learning	

#	Course Outcome	Description	Marks
1	1	CO 1	17
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	13
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025 ECE_5101H:Deep Learning	

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	13(26%)
3	Application	6	21(42%)
4	Analysis	5	16(32%)

5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)